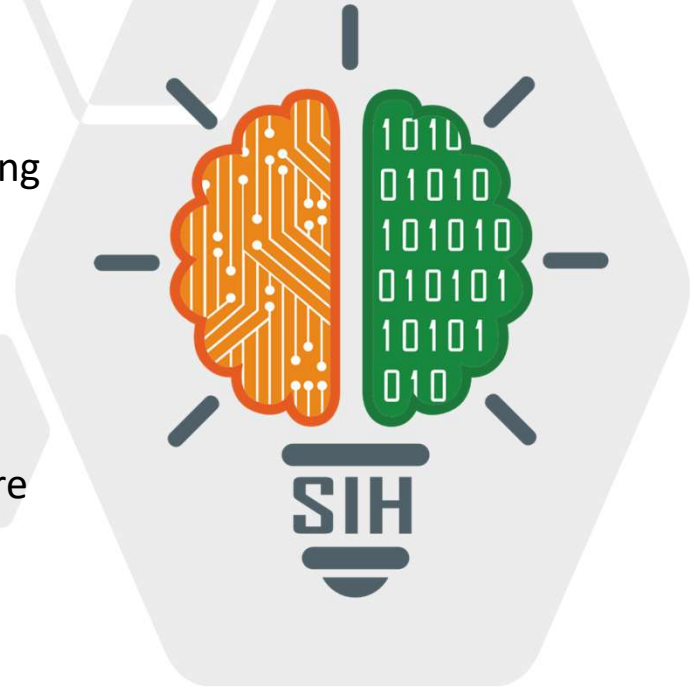
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- **Problem Statement ID -** 26232
- **Problem Statement Title -** Strengthening market linkages and price discovery for farmers
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category- Software/Hardware -** Software
- **Team ID-**
- **Team Name -** KisanX

Chanakya

FARM-TO-MARKET PRICE INTELLIGENCE & LINKAGE PLATFORM

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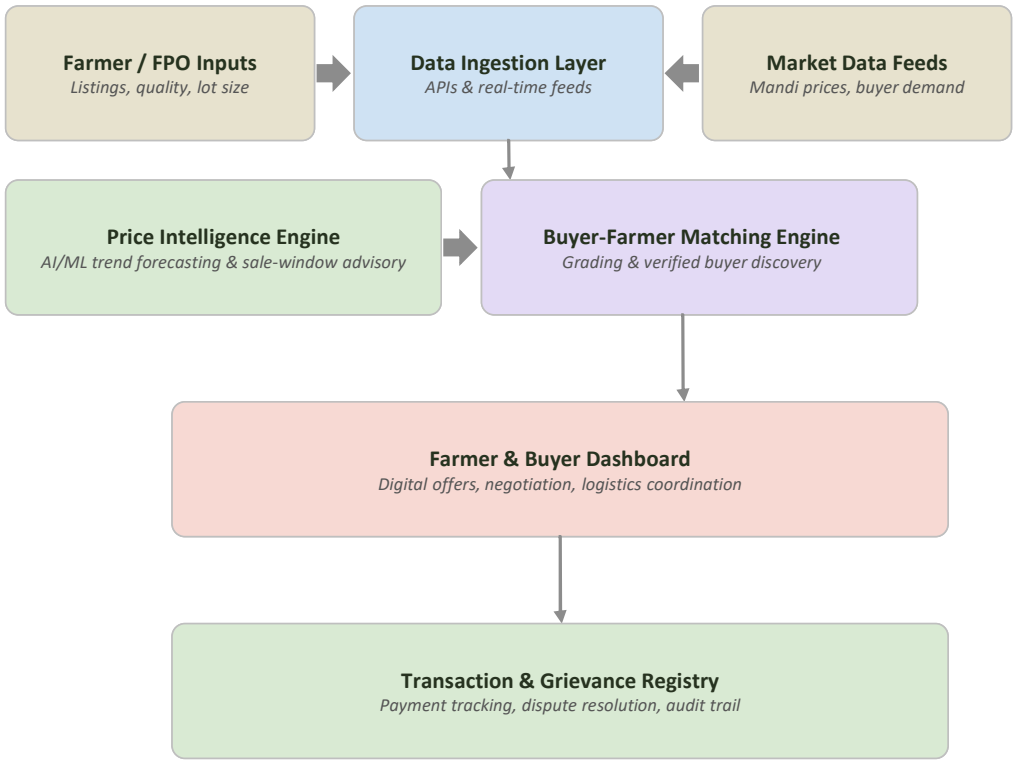
PROBLEM OVERVIEW

- ❑ Farmers have limited visibility into current & expected prices across mandis, processors, institutional buyers and digital trading channels.
- ❑ Quality specs, demand, logistics, storage, payment reliability and buyer credentials are fragmented and hard to verify.
- ❑ Liquidity and storage constraints push farmers into distress sales right after harvest, weakening bargaining power.
- ❑ Buyers struggle to aggregate consistent volumes and verify produce quality at scale.
- ❑ Net effect: opaque price discovery and unreliable farm-gate-to-buyer linkages.

PROPOSED SOLUTION

- ➢ Aggregate real-time mandi prices, buyer demand, arrival volumes, transport and storage options into one intelligence layer.
- ➢ Deliver localised price trends and AI-driven sale-window recommendations to farmers and FPOs.
- ➢ Match farmers/FPOs to verified buyers by quality, volume and logistics fit.
- ➢ Enable lot creation, quality grading, digital offers and logistics coordination end-to-end.
- ➢ Track payments digitally and support transparent dispute / grievance resolution.

ARCHITECTURE



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TECHNICAL APPROACH

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METHODOLOGY

Study farmer and FPO needs, price problems, and buyer verification requirements.

Requirement Analysis

Design a scalable system for price data, buyer matching, and dashboards.

System Architecture Design

Connect mandi APIs like eNAM/Agmarknet with weather, transport, and storage data.

Data Pipeline & API Integration

Build price prediction, buyer matching, and verification features.

Backend & AI/ML Development

Mobile App Development

Build easy multilingual farmer and buyer apps with offline support.

Security & Verification Hardening

Add KYC, buyer checks, and secure payment tracking.

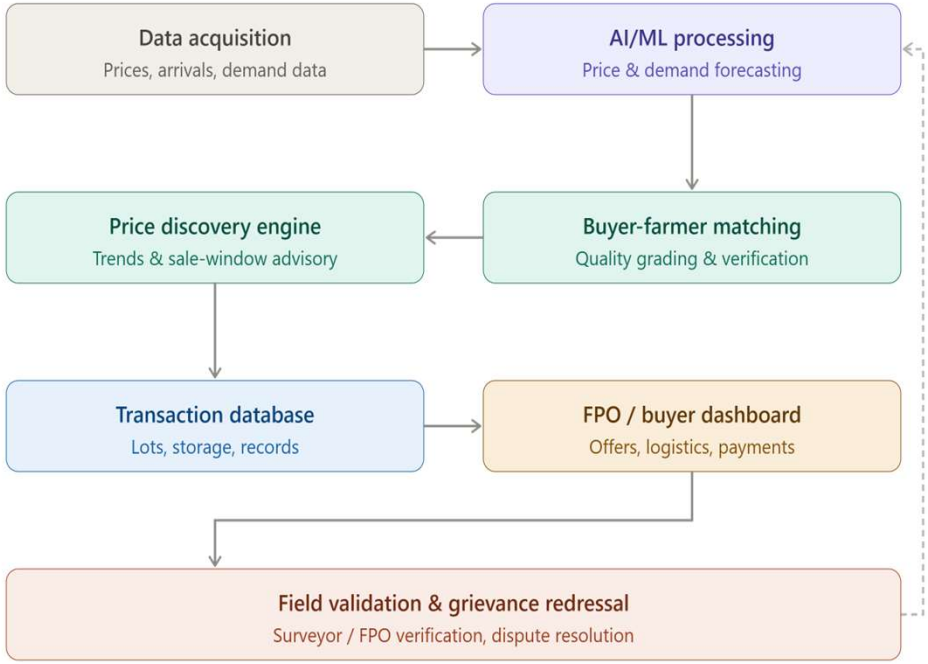
Pilot Deployment

Test the system with selected FPOs and mandis and collect feedback.

Full Rollout & Continuous Monitoring

Launch nationwide and keep improving the system through monitoring, training, and updates

FLOW CHART



TECH STACK

Python, django, HTML, CSS, SQLite, GitHub, Bootstrap, JS

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FEASIBILITY & VIABILITY

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FEASIBILITY

1. TECHNICAL FEASIBILITY

Proven technologies make the app **easy, practical, and low-risk** to build.

2. ECONOMICAL FEASIBILITY

Low running costs, supported by small fees and **FPO/government support**.

3. SOCIAL FEASIBILITY

Social Feasibility
Clear local-language prices help farmers **trust the app, get fair prices, and reduce middlemen.**

4. LEGAL FEASIBILITY

Follows **Indian data rules** and uses **official price sources**, making the platform **legal and government-friendly**.

5. OPERATIONAL FEASIBILITY

Simple local-language app, offline access, and admin support make it easy for farmers to use and manage.

6. SECURITY FEASIBILITY

Regular security checks, encrypted data, and buyer verification protect farmers from fraud, fake buyers, and payment problems.

POTENTIAL CHALLENGES AND RISKS

LOW DIGITAL LITERACY

Many farmers are not fully comfortable using a smartphone app or reading digital price charts.



FAKE OR UNVERIFIED BUYERS

Some buyers on the platform could be fraudulent, posting fake demand or attempting payment scams.



WEAK INTERNET IN VILLAGES

Poor network in remote areas can make it hard to see live prices at the right moment.



PRICE MANIPULATION

Incorrect mandi data or intentional price manipulation could mislead a farmer's selling decision.



STRATEGIES FOR OVERCOMING THESE CHALLENGES

SIMPLE UI & TRAINING

Easy, icon-based screens in local languages, plus training camps and voice-guided help.



BUYER VERIFICATION

Compulsory ID checks and a rating system so only verified, trustworthy buyers can transact.



OFFLINE-FIRST DESIGN

Prices are saved on the phone and updated automatically once internet return



DATA CHECKS & AUDITS

Prices are cross-checked from multiple official sources (eNAM, Agmarknet) with regular audits.



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IMPACT AND BENEFITS

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UNIQUE FEATURES:

Real-Time Price Aggregation

Mandi prices from multiple sources are pulled into one place, updated continuously.

AI Sale-Window Advisory

AI suggests the best time to sell based on price trends and demand.

End-to-End Lot Tracking

Every lot is tracked from listing to final payment, nothing lost in between.

Verified Buyer Network

Only ID-checked, rated buyers can place offers on the platform.

Offline-First Access

Prices and offers stay available even with poor or no internet.

IMPACTS

Fairer Prices for Farmers



Removes guesswork so farmers sell closer to the true market price, not a middleman's offer.

Reduced Distress Selling



Sale-window advice helps farmers avoid rushed, low-price sales right after harvest.

Trusted Buyer Access



Verified buyers replace unknown middlemen, cutting the risk of fraud.

Wider Market Reach



Farmers connect with buyers beyond their local mandi, not just nearby traders.

Reliable Farm Income



Steadier, better-informed sales support more predictable household income.

BENEFITS

Transparent Marketplace



One shared, trustworthy space for price discovery and trade.

Stronger FPOs



Farmer groups can aggregate produce and negotiate with more confidence.

Easy Price Tracking



Simple dashboards let farmers and buyers follow trends at a glance.

Scales Nationwide



Architecture supports growth from a few mandis to a national network.

Long-Term Value



Modest running costs deliver lasting gains in income and trust.

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RESEARCH AND REFERENCES

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- eNAM & Agmarknet (Government of India) portals for live mandi prices, arrival volumes and trading data.
- NABARD & SFAC guidelines on FPO formation, aggregation models and producer group financing.
- AGMARK / DMI quality-grading and specification standards for agricultural produce.
- AI/ML forecasting research (IEEE, Springer) for price prediction and demand estimation.
- NCCD cold-chain research on warehousing and logistics for reducing post-harvest storage loss.
- International best practices in agri-market linkage platforms (Ethiopia ECX, Kenya Twiga, Indonesia TaniHub).

Key References

eNAM
National Agriculture Market
Mandi Prices & Trading

Agmarknet
Govt. of India
Price & Arrival Data

NABARD / SFAC
FPO Guidelines
& Aggregation Support

IEEE Xplore
Research Papers
(AI/ML, Forecasting)

AGMARK / DMI
Quality Grading
& Certification Standards

World Bank / FAO
Global Best Practices
in Agri Market Linkage